Proposed Project Budget

Project: Benchmarking Visual Recovery from Incomplete Capture on Commodity Devices
Organization: FreeMoCap Foundation · **Period:** 12 months

Cash support requested: $80,000 · **AWS Promotional Credits requested:** $25,000

1. Cash budget

Cost category	Basis of estimate	Amount
Scientific Co-PI — Aaron T. Cherian	200 hours × $100/hour	$20,000
Technical Co-PI — Jonathan S. Matthis	200 hours × $100/hour	$20,000
Research Software Developer	425 hours × $80/hour	$34,000
Commodity-device testing	Computers, upgrades, cameras, and accessories	$4,000
Dissemination and travel	Research presentation and conference expenses	$2,000
Total cash request		**$80,000**

Personnel justification

Scientific Co-PI ($20,000). Lead benchmark study design, validation metrics, reference-data alignment, interpretation of performance/accuracy trade-offs, reproducibility checks, and scientific reporting (Aims 1–3).

Technical Co-PI ($20,000). Guide software architecture, integration of benchmark interfaces throughout FreeMoCap, technical review, experimental design, and release quality assurance (Aims 2–3).

Research Software Developer ($34,000). Implement hardware-aware inference and model benchmarking, including ONNX execution-provider and precision comparisons, multiview batching, method adapters, experiment automation, and reporting interfaces (primarily Aim 1, with contributions to Aims 2–3). This work may be performed through a direct project-services contract.

Other direct costs

Commodity-device testing ($4,000). Purchase or upgrade representative CPU-only, integrated-graphics, and consumer-GPU systems, as needed after inventorying existing equipment, with low-cost cameras, storage, and peripherals for repeatable local tests (Aims 1 and 3).

Dissemination and travel ($2,000). Support presenting findings, sharing benchmark methods, and engaging researchers at a relevant scientific or open-source conference (Aim 3).

2. AWS Promotional Credits

Planned use	Credits
EC2 GPU compute — model and recovery-method benchmark sweeps	$17,000
EC2 CPU compute — baselines, compatibility checks, regression testing	$4,000
S3 — permitted benchmark inputs and versioned results	$2,000
Supporting cloud usage — data transfer, logs, and related resources	$2,000
Total AWS credit request	**$25,000**

AWS Batch will coordinate repeatable jobs on EC2; S3 will hold permitted test inputs and outputs. These amounts are planning allocations, not claims about measured usage. Instance types, run counts, storage volumes, and transfer requirements will be refined from pilot benchmarks. Local device measurements will separately characterize actual on-device performance.

Budget basis: estimated direct project hours and rates, subject to final project agreements, plus itemized expenses. AWS credits are separate from the cash request; no AWS expenses are counted in both categories.